Cisco Introduction to Networks (Module 1)

1. Core Network Components

This section covers the fundamental building blocks of any network.

- **Hosts (End Devices):** All computers connected to a network that participate directly in communication. They are the source or destination of a message.
  - **Clients:** Hosts that request and display information.
  - **Servers:** Hosts that provide information or services to other hosts.
  - **Peer-to-Peer (P2P) Network:** A network where devices can function as both a client and a server simultaneously.
- **Intermediary Devices:** Devices that connect end devices to the network and can connect multiple networks to form an internetwork. They determine the path for messages.
  - **Examples:** Switches, Routers, Wireless Access Points, Firewalls.
- **Network Media:** The physical channel over which a message travels from source to destination.
  - **Types:**
    - **Copper Cable:** Uses electrical pulses.
    - **Fiber-Optic Cable:** Uses pulses of light.
    - **Wireless:** Uses electromagnetic waves.
- **Network Interface Card (NIC):** The physical hardware component that connects an end device to the network.
- **Port:** A specialized connection point on a networking device (like a switch or router) where a cable is plugged in to connect to a network segment.
- **Interface:** A specialized port on a networking device that connects to individual networks. Often used interchangeably with "port" in a software context (e.g., a router's interface to a LAN).

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2. Network Topologies

Topologies are the "blueprints" of a network, showing how devices are arranged and connected.

- **Physical Topology:** A diagram that shows the **actual physical location** of intermediary devices, end devices, and cable installation.
- **Logical Topology:** A diagram that shows how devices, ports, and the **addressing scheme** connect the network. It illustrates the flow of data, not the physical layout.

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3. Network Types & Infrastructures

Networks are categorized by their size, ownership, and purpose.

- **LAN (Local Area Network):** A network infrastructure that spans a **small geographical area**, such as a home, office, or department. It is typically owned and managed by a single organization or individual.
- **WAN (Wide Area Network):** A network infrastructure that spans a **large geographical area**, connecting multiple LANs. WANs are often owned and managed by a telecommunications service provider (ISP).
- **Intranet:** A **private network** (using LAN and WAN technologies) that belongs to an organization and is accessible only to its internal users (e.g., employees).
- **Extranet:** A network that provides **secure and safe access to individuals who work for a different organization** but require access to the organization's data (e.g., partners, suppliers, customers).

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4. Internet Connection Types

The choice of connection depends on the user's needs (home vs. business) and location.

For Home & Small Office (SOHO)

- **Cable:** High bandwidth, always-on connection over the same coaxial cable as cable TV.
- **DSL (Digital Subscriber Line):** High bandwidth, always-on connection that runs over a standard telephone line. (ADSL is common, with faster downloads than uploads).
- **Cellular:** Uses a mobile phone network. Provides access wherever there is a signal.
- **Satellite:** Useful in remote areas. Requires a clear line of sight to the satellite.
- **Dial-up Telephone:** An inexpensive but very low-bandwidth option that uses a phone line and modem.
- **Fiber Internet:** High-speed broadband using fiber-optic cables, offering symmetrical (equal) upload and download speeds.

For Businesses

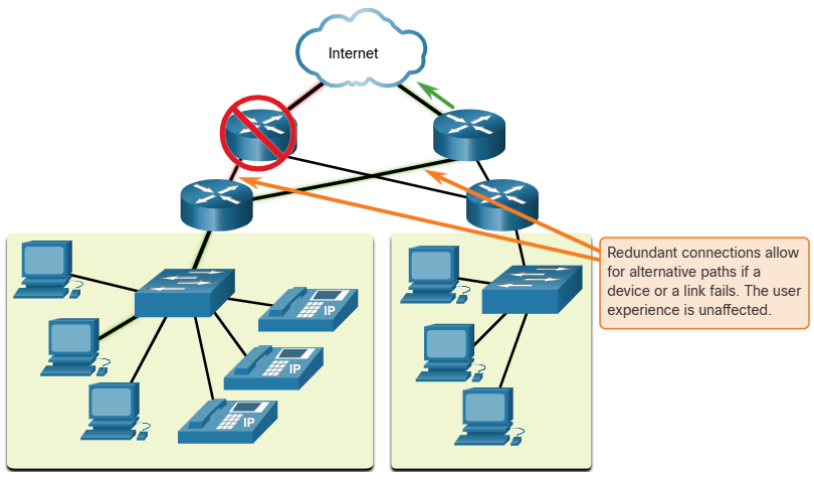
- **Dedicated Leased Line:** A reserved, private circuit within a service provider's network, rented monthly/yearly for guaranteed, reliable connectivity between offices.
- **Metro Ethernet (Ethernet WAN):** Extends LAN technology into the WAN, providing high-bandwidth connections between locations in a metropolitan area.
- **Business DSL (SDSL):** A Symmetric DSL service where upload and download speeds are equal.
- **Satellite & Fiber Internet:** Also used for business, with fiber being the preferred choice for high-performance needs.

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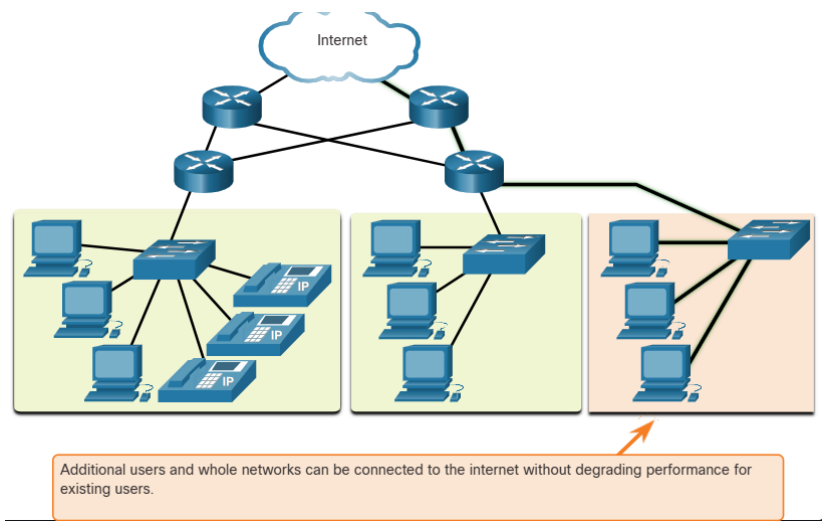
5. Key Network Architecture Characteristics

To meet user expectations, network architects must design networks with these four characteristics:

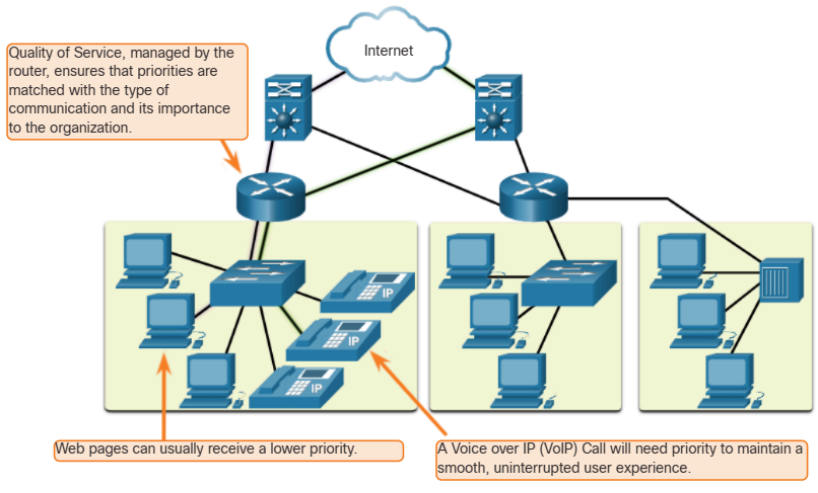
- **Fault Tolerance:** The ability of a network to limit the number of affected devices during a failure. This is often achieved through **redundancy** (having multiple paths to a destination).



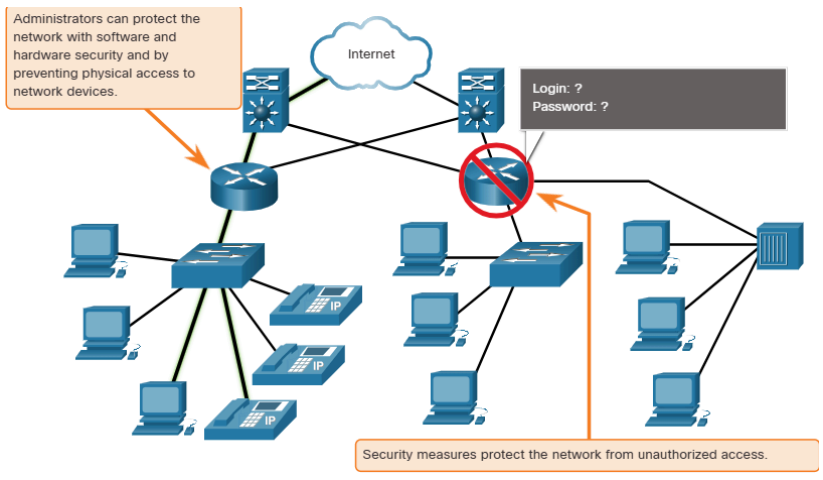
- **Scalability:** The ability of a network to expand quickly to support new users, devices, and applications without a significant drop in performance.



- **Quality of Service (QoS):** A mechanism for managing network congestion and ensuring the reliable delivery of critical content by prioritizing certain types of traffic (e.g., voice or video).



- **Security:** Protecting the network infrastructure and information. This is achieved through three primary goals (the CIA Triad):
  - **Confidentiality:** Only authorized individuals can access data.
  - **Integrity:** Data is not altered during transmission.
  - **Availability:** The network is always accessible to authorized users.



6. Converged Networks

- **Definition:** A single network infrastructure that carries multiple types of services—such as **data, voice, and video**—between many different types of devices.
- **Key Feature:** Unlike older dedicated networks, converged networks use the **same set of rules, agreements, and implementation standards** for all services, making them more efficient and cost-effective.

7. Network Trends & Technologies

These are recent developments that shape modern networking.

- **BYOD (Bring Your Own Device):** The policy of allowing employees to use their personal devices (laptops, phones, tablets) for work-related tasks on the company network.
- **Online Collaboration:** Tools (like Cisco Webex) that allow employees, students, and partners to connect and interact instantly.
- **Video Communications:** The use of video for collaboration, entertainment, and communication (e.g., video calls, streaming).
- **Cloud Computing:** The practice of storing and accessing data and applications over the internet instead of on a local hard drive. It includes:
  - **Public Clouds:** Services offered over the public internet (e.g., AWS, Google Cloud).
  - **Private Clouds:** Services offered over a private network to a single organization.
  - **Hybrid Clouds:** A combination of public and private clouds.
  - **Custom Clouds:** A cloud environment built to meet the specific needs of an organization.
- **Smart Home Technology:** The integration of networking technology into home appliances and devices for automation and remote control.
- **Powerline Networking:** A technology that sends data over existing electrical wiring in a building.
- **WISP (Wireless Internet Service Provider):** An ISP that provides internet access using wireless technologies like those found in home Wi-Fi networks.

8. Network Security Threats & Components

Common External Threats

- **Malware:** Viruses, worms, and Trojan horses.
- **Spyware & Adware:** Software that secretly gathers information or displays unwanted ads.
- **Zero-day Attacks:** Exploits for vulnerabilities that are unknown to the software vendor.
- **Threat Actor Attacks:** Targeted attacks from malicious individuals or groups.
- **Denial of Service (DoS):** An attack that overwhelms a network or service, making it unavailable to users.
- **Data Interception & Theft:** Unauthorized access to data in transit.
- **Identity Theft:** The fraudulent acquisition and use of a person's private identifying information.

Security Components

- **For Home & Small Office:**
  - **Antivirus & Antispyware:** Software to detect and remove malicious software.
  - **Firewall Filtering:** A basic firewall to block unauthorized access to the network.
- **For Larger & Corporate Networks:**
  - **Dedicated Firewall Systems:** More robust, hardware-based firewalls.

- **Access Control Lists (ACLs):** Rules that determine which traffic is allowed or denied.
- **Intrusion Prevention Systems (IPS):** Systems that actively monitor for and block malicious activity.
- **Virtual Private Networks (VPNs):** Encrypted tunnels that provide secure remote access to a network.